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Saeed Salehi

Associate Professor of Fluid Mechanics at Linköping University

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Current Position

Associate Professor Linköping University
2025 – present Division of Applied Thermodynamics and Fluid Mechanics, Department of Management and Engineering. My research focuses on combining data-driven methods and artificial intelligence with Computational Fluid Dynamics (CFD) to develop efficient and reliable methods for simulation, model reduction, control, and uncertainty quantification of fluid flows.

Previous Appointments

Researcher Employed by Chalmers Industriteknik (CIT), conducting research on
2023 – 2025 the application of artificial intelligence and machine learning in computational fluid dynamics. During my time at CIT, I was hired by Chalmers University of Technology for a project on "Artificial Intelligence in Fluid Dynamics for Improving the Lifetime of Hydraulic Turbines".

Postdoc Chalmers University of Technology
2019 – 2023 Project: New operating procedures of hydropower for a sustainable energy system. During my postdoctoral position, my research focused on developing advanced numerical methods in OpenFOAM to simulate the transient operations of hydraulic turbines. I also expanded my expertise in data-driven and machine-learning approaches, particularly in the field of reduced-order modeling, such as Proper Orthogonal Decomposition (POD) and Dynamic Mode Decomposition (DMD). The research resulted in several publications in high-impact, peer-reviewed journals and an open-source software package for simulating hydraulic turbine transients, which is being utilized by Vattenfall, a leading energy company in Sweden.

Lecturer University of Science and Culture, Tehran, Iran
2018 – 2019 Taught "Introduction to Turbomachinery" at the undergraduate level

Education

PhD
2012 – 2018

University of Tehran, Mechanical Engineering, Hydraulic Machinery
Thesis: Uncertainty quantification of turbulent flows in hydraulic machinery
Supervisors: Prof. Mehrdad Raisee, Prof. Michel Cervantes, Prof. Ahmad Nourbakhsh
The thesis investigates the impact of operational and geometrical uncertainties on the performance of hydraulic machinery, extending uncertainty quantification (UQ) to this critical field using polynomial chaos expansion (PCE). To address the high computational costs of traditional UQ methods, I developed efficient techniques such as compressed sensing and multifidelity ℓ_1 minimization, validated through both analytical and CFD problems. Real-world case studies demonstrate that combined operational and geometrical uncertainties significantly influence the flow and performance of hydraulic pumps. This work advances UQ methods, contributing to more robust and reliable hydraulic machinery design.

MSc.
2010 – 2012

University of Tehran, Mechanical Engineering, Energy Conversion
Thesis: Computation of steady and pulsating turbulent flow through a straight asymmetric diffuser with moderate adverse pressure gradient
Supervisors: Prof. Mehrdad Raisee, Prof. Michel Cervantes

BSc.
2006 – 2010

University of Tehran, Mechanical Engineering
Thesis: Large eddy simulation of stall development around airfoils

Research Grants

- **Enabling Flexible Hydropower Operation by AI-Based Active Flow Control**
  Funding agency: ÅForsk Foundation
  Time period: from 2027 to 2031 (5 years)
  Amount: 4 000 000 SEK
  Role: PI
  Description: Awarded by the ÅForsk Foundation through their Early-Career Research Grant program. The project combines CFD, reinforcement learning-based active flow control, and hydropower to enable more flexible hydropower operation. A PhD student will be recruited at Linköping University. The total project budget including departmental co-funding is 5 550 000 SEK.

- **Artificial Intelligence for Enhanced Hydraulic Turbine Lifetime**
  Funding agency: Swedish Hydropower Centre (SVC)
  Time period: from 2023-01 to 2027-06 (four and half years)
  Amount: 7 411 000 SEK
  Role: Co-PI (PI: Prof. Håkan Nilsson)
  Description: This is the successful grant that I received for my current position as a researcher at Chalmers Industriteknik and Chalmers University of Technology. I developed the idea for the project and had the leading role in writing the grant application. A requirement for the call is also that the university provides 39.5% in-kind to the project. Therefore, the grant also includes funding for a PhD student (as in-kind) that we recently hired and the total budget sums up to 12 498 000 SEK.

- **Hydropower Operation and Lifetime Analysis**
  Funding agency: Swedish Hydropower Centre (SVC)
  Time period: from 2023-03 to 2027-08 (four and half years)
  Amount: 5 266 000 SEK
  Role: Co-PI (PI: Prof. Håkan Nilsson)
  Description: The funding for this successful grant application was used to hire a PhD student (Martina) whom I am co-supervising. I had a major role in developing the application as it can be considered a continuation of my postdoc project.

- **Multi-Fidelity Physics-Informed Neural Network to Solve Partial Differential Equations**
  Funding agency: Chalmers University of Technology (internal call, in competition)
  Time period: from 2023-01 to 2027-06 (four and half years)
  Amount: 5 517 000 SEK
  Role: Co-PI (A collaborative application, i.e., no PI or main applicant)
  Description: This is the funding that we received for a cross-divisional PhD student in the Department of Mechanics and Maritime Sciences of Chalmers University of Technology. I played a pivotal role in formulating the concept and preparing the grant proposal application. Subsequently, the department approved our proposal, leading to the recruitment of a PhD student (Mohammad), whom I am co-supervising.

Publications

Peer-reviewed journal papers

The list of publications is presented in reverse chronological order.

[1] Martina Nobilo, **Saeed Salehi**, and Håkan Nilsson. "Lifetime analysis of hydro turbines with focus on fatigue damage in a renewable energy system – A review". *Renewable and Sustainable Energy Reviews* 228 (2026), p. 116578. DOI: 10.1016/j.rser.2025.116578.

[2] **Saeed Salehi**. "Transfer learning strategies for accelerating reinforcement-learning-based flow control" (2025). arXiv: 2510.16016 [cs.LG].

[3] Mohammad Sheikholeslami, **Saeed Salehi**, Wengang Mao, Arash Eslamdoost, and Håkan Nilsson. "Physics-informed neural networks with hard and soft boundary conditions for linear free surface waves". *Physics of Fluids* 37.8 (2025), p. 087158. DOI: 10.1063/5.0277421.

[4] **Saeed Salehi**. "An efficient intrusive deep reinforcement learning framework for OpenFOAM". *Meccanica* 60.6 (2025), pp. 1673–1693. DOI: 10.1007/s11012-024-01830-1.

[5] Jonathan Fahlbeck, Håkan Nilsson, Mohammad Hossein Arabnejad, and **Saeed Salehi**. "Performance characteristics of a contra-rotating pump-turbine in turbine and pump modes under cavitating flow conditions". *Renewable Energy* (2024). DOI: 10.1016/j.renene.2024.121605.

[6] Faiz Azhar Masoodi, **Saeed Salehi**, and Rahul Goyal. "Reorganization of flow field due to load rejection driven self-mitigation of high load vortex breakdown in a Francis turbine". *Physics of Fluids* 36.9 (2024), p. 094110. DOI: 10.1063/5.0222739.

[7] **Saeed Salehi** and Håkan Nilsson. "Modal analysis of vortex rope using dynamic mode decomposition". *Physics of Fluids* 36.2 (2024), p. 024122. DOI: 10.1063/5.0186871.

[8] Faiz Azhar Masoodi, **Saeed Salehi**, and Rahul Goyal. "Formation and evolution of vortex breakdown consequent to post design flow increase in a Francis turbine". *Physics of Fluids* 36.2 (2024), p. 025116. DOI: 10.1063/5.0187104.

[9] **Saeed Salehi** and Håkan Nilsson. "A semi-implicit slip algorithm for mesh deformation in complex geometries, implemented in OpenFOAM". *Computer Physics Communications* 287 (2023), p. 108703. DOI: 10.1016/j.cpc.2023.108703.

[10] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "Surrogate based optimisation of a pump mode startup sequence for a contra-rotating pump-turbine using a genetic algorithm and computational fluid dynamics". *Journal of Energy Storage* 62 (2023), p. 106902. DOI: 10.1016/j.est.2023.106902.

[11] **Saeed Salehi** and Håkan Nilsson. "Effects of uncertainties in positioning of PIV plane on validation of CFD results of a high-head Francis turbine model". *Renewable Energy* 193 (2022), pp. 57–75. DOI: 10.1016/j.renene.2022.04.018.

[12] **Saeed Salehi** and Håkan Nilsson. "Flow-induced pulsations in Francis turbines during startup - A consequence of an intermittent energy system". *Renewable Energy* (2022). DOI: 10.1016/j.renene.2022.01.111.

[13] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "A head loss pressure boundary condition for hydraulic systems". *OpenFOAM Journal* 2 (2022), pp. 1–12. DOI: 10.51560/ofj.v2.69.

[14] **Saeed Salehi**, Håkan Nilsson, Eric Lillberg, and Nicolas Edh. "An in-depth numerical analysis of transient flow field in a Francis turbine during shutdown". *Renewable Energy* 179 (2021), pp. 2322–2347. DOI: 10.1016/j.renene.2021.07.107.

[15] **Saeed Salehi** and Håkan Nilsson. "OpenFOAM for Francis turbine transients". *OpenFOAM Journal* 1 (2021), pp. 47–61. DOI: 10.51560/ofj.v1.26.

[16] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "Flow Characteristics of Preliminary Shutdown and Startup Sequences for a Model Counter-Rotating Pump-Turbine". *Energies* 14.12 (2021). DOI: 10.3390/en14123593.

[17] Mohamad Sadeq Karimi, Mehrdad Raisee, **Saeed Salehi**, Patrick Hendrick, and Ahmad Nourbakhsh. "Robust optimization of the NASA C3X gas turbine vane under uncertain operational conditions". *International Journal of Heat and Mass Transfer* 164 (2021), p. 120537. DOI: 10.1016/j.ijheatmasstransfer.2020.120537.

[18] Akbar Mohammadi Ahmar, **Saeed Salehi**, and Mehrdad Raisee. "Uncertainty quantification of the turbulent flow field and heat transfer of film cooling (in Persian)". *Journal of Solid and Fluid Mechanics* 10.2 (2020), pp. 177–192.

[19] Mohamad Sadeq Karimi, **Saeed Salehi**, Mehrdad Raisee, Patrick Hendrick, and Ahmad Nourbakhsh. "Probabilistic CFD computations of gas turbine vane under uncertain operational conditions". *Applied Thermal Engineering* 148 (2019), pp. 754–767. DOI: 10.1016/j.applthermaleng.2018.11.072.

[20] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "On the flow field and performance of a centrifugal pump under operational and geometrical uncertainties". *Applied Mathematical Modelling* 61 (2018), pp. 540–560. DOI: 10.1016/j.apm.2018.05.008.

[21] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "An efficient multifidelity ℓ_1-minimization method for sparse polynomial chaos". *Computer Methods in Applied Mechanics and Engineering* 334 (2018), pp. 183–207. DOI: 10.1016/j.cma.2018.01.055.

[22] Ali Salehpour, **Saeed Salehi**, Samaneh Salehpour, and Mehdi Ashjaee. "Thermal and hydrodynamic performances of MHD ferrofluid flow inside a porous channel". *Experimental Thermal and Fluid Science* 90 (2018), pp. 1–13. DOI: 10.1016/j.expthermflusci.2017.08.032.

[23] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "Efficient uncertainty quantification of stochastic CFD problems using sparse polynomial chaos and compressed sensing". *Computers & Fluids* 154 (2017), pp. 296–321. DOI: 10.1016/j.compfluid.2017.06.016.

[24] **Saeed Salehi**, Mehrdad Raisee, Michel J. Cervantes, and Ahmad Nourbakhsh. "The Effects of Inflow Uncertainties on the Characteristics of Developing Turbulent Flow in Rectangular Pipe and Asymmetric Diffuser". *Journal of Fluids Engineering* 139.4 (2017), p. 041402. DOI: 10.1115/1.4035302.

[25] **Saeed Salehi**, Mehrdad Raisee, and Michel J. Cervantes. "Computation of Developing Turbulent Flow Through a Straight Asymmetric Diffuser With Moderate Adverse Pressure Gradient". *Journal of Applied Fluid Mechanics* 10.4 (2017), pp. 1029–1043. DOI: 10.18869/acadpub.jafm.73.241.26311.

[26] **Saeed Salehi** and Mehrdad Raisee. "Application of Gram-Schmidt orthogonalization method in uncertainty quantification of computational fluid dynamics problems with arbitrary probability distribution functions (In Persian)". *Modares Mechanical Engineering* 15.12 (2015), pp. 1–8.

[27] Behrouz Takabi and **Saeed Salehi**. "Augmentation of the Heat Transfer Performance of a Sinusoidal Corrugated Enclosure by Employing Hybrid Nanofluid". *Advances in Mechanical Engineering* 6 (2014), p. 147059. DOI: 10.1155/2014/147059.

Book chapters

[1] **Saeed Salehi**, Mehrdad Raisee, Michel Cervantes, and Ahmad Nourbakhsh. "Development of an Efficient Multifidelity Non-Intrusive Uncertainty Quantification Method". *Evolutionary and Deterministic Methods for Design Optimization and Control With Applications to Industrial and Societal Problems*. Ed. by Esther Andrés-Pérez, Leo M. González, Jacques Periaux, Nicolas Gauger, Domenico Quagliarella, and Kyriakos Giannakoglou. Springer International Publishing, 2019. DOI: 10.1007/978-3-319-89890-2.

Conference proceedings

[1] Martina Nobilo, Saeed Salehi, and Håkan Nilsson. "On the flow-induced pulsating forces during load reduction of a Kaplan turbine model". Vol. 1483. 1. IOP Publishing, 2025, p. 012022. DOI: 10.1088/1755-1315/1483/1/012022.

[2] J Fahlbeck, H Nilsson, and S Salehi. "On the pump mode shutdown sequence for a model contra-rotating pump-turbine". Vol. 1483. 1. IOP Publishing, 2025, p. 012016. DOI: 10.1088/1755-1315/1483/1/012016.

[3] **Saeed Salehi** and Håkan Nilsson. "Dynamic Mode Decomposition of Rotating Vortex Rope Instability". *9th IAHR Meeting of the WorkGroup on Cavitation and Dynamic Problems in Hydraulic Machinery and Systems*. 2024.

[4] Martina Nobilo, Saeed Salehi, and Håkan Nilsson. "Effects of load reduction on forces and moments on the runner blades of a Kaplan turbine model". Vol. 1411. 1. IOP Publishing, 2024, p. 012001. DOI: 10.1088/1755-1315/1411/1/012001.

[5] J Fahlbeck, H Nilsson, and S Salehi. "Analysis of mode-switching of a contra-rotating pump-turbine based on load gradient limiting shutdown and startup sequences". Vol. 1411. 1. IOP Publishing, 2024, p. 012049. DOI: 10.1088/1755-1315/1411/1/012049.

[6] Mohammad Sheikholeslami, **Saeed Salehi**, Wengang Mao, Arash Eslamdoost, and Håkan Nilsson. "Physics-Informed Neural Networks for Modeling Linear Waves". Vol. Volume 9: Philip Liu Honoring Symposium on Water Wave Mechanics and Hydrodynamics; Blue Economy Symposium. International Conference on Offshore Mechanics and Arctic Engineering. 2024, V009T12A002. DOI: 10.1115/OMAE2024-125048.

[7] Jonathan Fahlbeck, Håkan Nilsson, and **Saeed Salehi**. "Evaluation of startup time for a model contra-rotating pump-turbine in pump-mode". Vol. 1079. 1. IOP Publishing, 2022, p. 012034. DOI: 10.1088/1755-1315/1079/1/012034.

[8] **Saeed Salehi**, Håkan Nilsson, Eric Lillberg, and Nicolas Edh. "Development of a novel numerical framework in OpenFOAM to simulate Kaplan turbine transients". Vol. 774. 1. IOP Publishing, 2021, p. 012058. DOI: 10.1088/1755-1315/774/1/012058.

[9] **Saeed Salehi**, Håkan Nilsson, Eric Lillberg, and Nicolas Edh. "Numerical Simulation of Hydraulic Turbine During Transient Operation Using OpenFOAM". Vol. 774. 1. IOP Publishing, 2021, p. 012060. DOI: 10.1088/1755-1315/774/1/012060.

[10] Jonathan Fahlbeck, Håkan Nilsson, **Saeed Salehi**, Mehrdad Zangeneh, and Melvin Joseph. "Numerical analysis of an initial design of a counter-rotating pump-turbine". Vol. 774. 1. IOP Publishing, 2021, p. 012066. DOI: 10.1088/1755-1315/774/1/012066.

[11] Mohamad Sadeq Karimi, **Saeed Salehi**, Mehrdad Raisee, and Ahmad Nourbakhsh. "Conjugate Heat Transfer Simulation of a Cooled Turbine Vane under Uncertain Operational Condition". *International Conference on Evolutionary and Deterministic Methods for Design Optimization and Control with Applications to Industrial and Societal Problems, Madrid, Spain.* 2017.

[12] **Saeed Salehi**, Mehrdad Raisee, Michel Cervantes, and Ahmad Nourbakhsh. "Development of an Efficient Multifidelity Non-Intrusive Uncertainty Quantification Method". *International Conference on Evolutionary and Deterministic Methods for Design Optimization and Control with Applications to Industrial and Societal Problems, Madrid, Spain.* 2017.

[13] **Saeed Salehi**, Mehrdad Raisee, and Ahmad Nourbakhsh. "Effects of Geometrical and Operational Uncertainties on the Performance of Hydraulic Machinery". *The 14th Asian International Conference on Fluid Machinery, Zhenjiang, China.* 2017.

[14] Vahid Etemadeasl, **Saeed Salehi**, Mehrdad Raisee, and Ahmad Nourbakhsh. "Numerical Investigation of Turbulent Flow in Francis-99 Turbine Using Various Turbulence Models". *The 14th Asian International Conference on Fluid Machinery, Zhenjiang, China.* 2017.

[15] Ehsan Akbari, **Saeed Salehi**, and M. Karami. "Numerical and Experimental Investigation of the Effect of Heat Exchanger Shape on Performance of an Industrial Heater (In Persian)". *21st Annual International Conference on Mechanical Engineering (ISME), Tehran, Iran.* 2013.

[16] Behrouz Takabi, **Saeed Salehi**, and Mohammad Hassan Rahimyan. "Studying the effects of employing hybrid nanofluid on heat transfer performance of a wavy cavity". *21st Annual International Conference on Mechanical Engineering (ISME), Tehran, Iran.* 2013.

Conference presentations

[1] **Saeed Salehi**. "Closed-loop control of vortex rope in a swirl generator using deep reinforcement learning". 2026.

[2] **Saeed Salehi**. "CFD with OpenSource Software: a case study of project-based learning in an international PhD course". 2024.

[3] **Saeed Salehi**. "Towards practical applications of deep reinforcement learning in CFD". 2024.

[4] **Saeed Salehi**. "Implementation of deep reinforcement learning in OpenFOAM for active flow control". 2023.

[5] **Saeed Salehi**. "Deep reinforcement learning for active flow control". 2023.

[6] **Saeed Salehi**. "Evolution of flow features during transient operation of a Kaplan turbine". 2022.

[7] **Saeed Salehi**. "Numerical simulation of hydraulic turbine during transient operation using OpenFOAM". 2020.

[8] **Saeed Salehi**. "Quantification of epistemic uncertainties in the $k - \varepsilon$ model coefficients". 2019.

Supervision Experience

I am supervising or have supervised the following students.

- **Jonathan Fahlbeck**, PhD student, Chalmers University of Technology, 2020 – 2024 (graduated)
  Project: Flow in contra-rotating pump-turbines at stationary, transient, and cavitating conditions (ALPHEUS project – European Union's Horizon 2020 program)
  Role: Co-supervisor

- **Martina Nobilo**, PhD student, Chalmers University of Technology, 2023 – 2025 (graduated with Licentiate degree)
  Project: CFD for hydropower lifetime analysis
  Role: Co-supervisor

- **Mohammad Sheikholeslami**, PhD student, Chalmers University of Technology, 2023 – present
  Project: Multi-Fidelity Physics-Informed Neural Network to solve partial differential equations
  Role: Co-supervisor

Teaching Experience

This summary outlines my primary teaching experiences in academia, showcasing my contributions to undergraduate and graduate education. For comprehensive information about my academic teaching, as well as my broader teaching experiences beyond academia, please refer to my pedagogical portfolio.

- **Introduction to turbomachinery**
  Role: Main teacher and examiner
  Place: University of Science and Culture, Iran
  Level: Bachelor
  Year: 2018 – 2019
  Credits: 7.5 ECTS (converted)
  Sessions: 32 sessions (48 hours total)
  No. of passed students per year: 20

- **Introduction to fluid dynamics in water turbines**
  Role: Main teacher and examiner
  Place: SVC research school at KTH Royal Institute of Technology
  Level: PhD
  Year: 2022
  Credits: 2.0 ECTS
  Sessions: 10 sessions (18 hours total)
  No. of passed students per year: 5

- **CFD with OpenSource Software**
  Role: Lecturer, Teaching assistant
  Place: Chalmers University of Technology
  Level: PhD
  Year: 2020 –
  Credits: 7.5 ECTS
  Sessions: Flipped classroom format
  No. of passed students per year: 7 – 13

- **Basic Usage of OpenFOAM**
  Role: Lecturer, Teaching assistant
  Place: Chalmers University of Technology
  Level: PhD
  Year: 2021 –
  Credits: 2.0 ECTS
  Sessions: Flipped classroom format
  No. of passed students per year: 17 – 30

- **Teaching assistant** (Turbomachinery Lab., Turbulent Flows, Introduction to Heat Transfer, Introduction to Fluid Mechanics)
  Place: University of Tehran, Iran
  Level: Bachelor and Master
  Year: 2013 – 2017

Pedagogical courses

In October 2023, I was awarded the “Diploma in Teaching and Learning in Higher Education" in recognition of the successful fulfillment of all program requirements, including seven courses. The Diploma is designed to meet the learning outcomes defined by the Association of Swedish Higher Education Institutions (SUHF). The program is equivalent to 15 ECTS credits or 10 weeks of full-time study. It concludes with conducting a final pedagogical project related to the teaching activities. More information about my final project is provided in my pedagogical portfolio.

Additionally, to further refine my abilities in the realm of supervision, I completed the “Supervising Research Students" course (3.0 ECTS) which is not included in the teaching diploma program. The following table presents a summary of my pedagogical training.

For more details, please see my pedagogical portfolio.

Overview of the completed courses related to pedagogical development

No.	Course name	Course code	Credits	Year
1	University teaching and learning	CLS925	2.5 ECTS	2021
2	Diversity and inclusion for learning in higher education	CLS930	2.0 ECTS	2023
3	Supervising writing processes	CLS910	2.5 ECTS	2023
4	Theoretical perspectives on learning	CLS900	2.5 ECTS	2023
5	Enhancing learning through writing	CLS941	4.5 ECTS	2023
6	Minor independent study in teaching and learning	CLS946	0.5 ECTS	2023
7	Reflections on teaching and learning in higher education	CLS920	0.5 ECTS	2023
8	Supervising research students	CLS905	3.0 ECTS	2022
	Total		**18.0 ECTS**	

Academic Service

Editorial roles **Section Editor**, OpenFOAM Journal, 2026 – present

Committees **Technical Committee Member**, OpenFOAM Turbomachinery Technical Committee, OpenFOAM Governance, 2022 – present

PhD examination committees

External Reviewer (90% seminar), University of Gävle, Sweden, 2026
Thesis: “Experimental and Numerical Evaluation of Wall-Confluent Jets Thermal Performance for Greenhouse Climate Control”. PhD candidate: Gasper Choonya

External Examiner (Defense committee), Universitat Rovira i Virgili, Tarragona, Spain, 2024
Thesis: “Numerical Modeling of High Aspect Ratio Fibers in Fluid Flows”. PhD candidate: MohammadJavad Norouzi

Reviewer Physics of Fluids (*AIP*), AIAA Journal (*AIAA*), Computers and Fluids (*Elsevier*), Applied Mathematical Modeling (*Elsevier*), Journal of Fluids Engineering (*ASME*), Biomechanics and Modeling in Mechanobiology (*Springer*), Science Progress (*Sage*), Fluids (*MDPI*), Energies (*MDPI*), Algorithms (*MDPI*), Processes (*MDPI*), Applied Fluid Mechanics (*IUT*)

Invited Speaker

- Swedish Society for Industrial Flow and Heat Transfer, May 2026, Lund, Sweden

  Title: Towards efficient control of turbulence

- LKAB Company, February 2026, Kiruna, Sweden (held online)

  Title: Engineering applications of computational and data-driven fluid dynamics

- Swedish Society for Industrial Flow and Heat Transfer, May 2024, Finspång, Sweden

  Title: Artificial intelligence and data-driven algorithms for flow feature extraction and control

- Dutch OpenFOAM User group meetings, July 2023

  Title: Deep Reinforcement Learning in CFD, Implementation in OpenFOAM

- OpenFOAM Student Workshop at Delft University of Technology, January 2023

  Role: Invited OpenFOAM expert; provided guidance on advanced OpenFOAM applications for turbomachinery and answered questions from PhD researchers

Utilization of My Research

- During my postdoc, I developed the required numerical methods to study transients in hydraulic turbines and published the framework as an open-source package (`semiImplicitSlip`). The Research and Development department of Vattenfall, a leading energy production company in Europe, is now using my development to study the transient operation of their hydropower systems.

- I also have other open-source GitHub projects aimed at making my research beneficial for both the academic and industrial communities. For more information, please refer to my GitHub at github.com/salehisaeed.

Open-Source Software

- `TensorforceFoam`: A TensorFlow-based intrusive Deep Reinforcement Learning (DRL) framework for OpenFOAM. The Tensorforce package is utilized for the DRL computations. The term intrusive refers to the direct integration of the DRL agent within the CFD environment. Such integration eliminates the need for any external information exchange during DRL episodes. The framework is parallelized using the message passing interface (MPI) for Python (mpi4py) to manage parallel environments for computationally intensive CFD cases through distributed computing.

- `semiImplicitSlip`: A mesh motion library in OpenFOAM that implements a robust semi-implicit algorithm for slipping the mesh points on curved surfaces. The algorithm includes two steps, namely, an explicit step based on the general slip condition and an implicit step based on the Dirichlet condition. It employs the Laplacian smoothing equations to spread the mesh deformation into the domain. Additionally, a solid-body rotation may be added on top of the deformed mesh, which could be useful for modelling the runner region in transient operation of Kaplan turbines which contains simultaneous mesh deformation and solid-body rotation of the mesh.

- `Francis-99 transients`: Open-source case and code for CFD simulation of the transient operation of a Francis turbine in OpenFOAM.

Computer Skills

CFD: OpenFOAM, Fluent, CFX

Programming: C++, Python, MATLAB, LaTeX

AI / ML: TensorFlow, PyTorch

Post-processing: ParaView, Tecplot

General: Linux, Git

Languages

- English: Fluent
- Swedish: Fluent
- Persian: Mother tongue